

# Underwater Image Enhancement

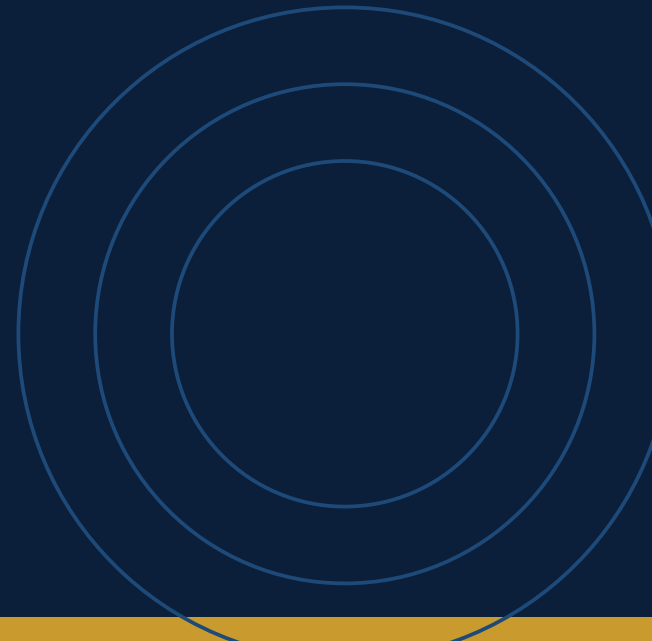
## Problem Statement & Proposed Solution

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Presented by

**Teja Sree Pasapula**

B.Tech, Computer Science and Engineering  
Mohan Babu University



# Problem Statement & Proposed Solution

## Problem Statement

- Color distortion
- Light scattering / haze
- Low contrast
- Severe blurring

### Core question:

*How can we improve underwater image visibility without losing important structural details and object boundaries?*

### Scope:

Software-only solution — no hardware integration or physical data collection.

## How We Solve It

### 1. Input

Paired underwater dataset (degraded + reference images)

### 2. Preprocessing

Preprocessing, EDA, and Train / Validation / Test split

### 3. Enhancement

Classical baseline vs. supervised image-to-image model

### 4. Edge Preservation

Check whether object boundaries are retained

### 5. Evaluation

PSNR, SSIM, UIQM / UCIQE, Edge-Preservation Index, Processing Time

# Proposed Pipeline

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